

$$XY = 26$$

$$X + Y = 8$$

$$x \times y = 12$$

true value = 100.0 units

= 102.6 units

$$= 102.6 - 100.0$$

$$= 2.6$$

100.0

$$R_s = 0.0266 \Omega \pm 0.25\%$$

$$P = I^2 R$$

$$P = IR(I)$$

$$P = VI \quad \text{--- (i)}$$

$$V = IR$$

$$I = \frac{V}{R} \quad \text{--- (2)}$$

$$W_{R_5} = 0.0266 \times 0.0025$$

$$= 6.65 \times 10^{-5}$$

sub ② into ①

$$P = V \left(\frac{V}{R} \right)$$

$$= 10 \left(\frac{1.2}{0.0286} \right)$$

$$= 451.128 \text{ W}$$

$$\frac{W_R}{R} = \sqrt{\left(\frac{a_E W_E}{E}\right)^2 + \left(\frac{a_{E_s} W_{E_s}}{E_s}\right)^2 + \left(\frac{a_{R_s} W_{R_s}}{R_s}\right)^2}$$

$$= \sqrt{\left(\frac{(1)(0.1)}{10}\right)^2 + \left(\frac{(1)(0.026)}{1.2}\right)^2 + \left(\frac{(-1)(6.65 \times 10^5)}{0.0266}\right)^2}$$

$$\frac{W_R}{R} = 0.02399$$

$$w_R = 0.02399(451.128)$$

$$W_R = 10.8226 \text{ W}$$

$$\therefore P = 451.128 \pm 10.82 \text{ W}$$

Assignment 3

$$\begin{array}{ccc}
 R_1 & & R_2 \\
 \text{---} \text{---} \text{---} & & \text{---} \text{---} \text{---} \\
 R_1 = (100 + (x+y) \times 2) \Omega & & R_2 = (50 + (x+y) \times 5) \Omega \\
 = (100 + 8 \times 2) \Omega & & = (50 + 8 \times 5) \Omega \\
 R_1 = 116 \Omega \pm 5\% & & R_2 = 90 \Omega \pm 5\%
 \end{array}$$

$$\begin{aligned}
 R_{\text{total}} &= R_1 + R_2 \\
 &= 116 + 90 \\
 &= 206 \Omega
 \end{aligned}$$

$$\begin{aligned}
 \omega_R &= \sqrt{(116 \times 0.05)^2 + (90 \times 0.05)^2} \\
 \omega_R &= 7.34 \Omega
 \end{aligned}$$

$$\therefore R_{\text{total}} = 206 \pm 7.34 \Omega$$

Assignment 4

$$\begin{aligned}
 L_1 &= 10.0 + 0.2 = 10.2 \\
 L_2 &= 10.2 + 0.6 = 10.8 \\
 L_3 &= 9.8 + 0.2 = 10.0 \\
 L_4 &= 10.1 + 0.6 = 10.7
 \end{aligned}$$

$$\bar{x} = \frac{10.2 + 10.8 + 10.0 + 10.7}{4} = 10.425$$

	x	$x - \bar{x}$	$(x - \bar{x})^2$
L_1	10.2	-0.225	0.0506
L_2	10.8	0.375	0.1406
L_3	10.0	-0.425	0.1806
L_4	10.7	0.275	0.0756

$$\begin{aligned}
 \sigma &= \sqrt{\frac{0.0506 + 0.1406 + 0.1806 + 0.0756}{4 - 1}} \\
 &= 0.3862
 \end{aligned}$$

Assignment 5

$$R_0 = (100 + 26) \Omega \pm 1\%, \quad 20^\circ\text{C}$$

$$= 126 \Omega \pm 1\%, \quad 20^\circ\text{C}$$

$$\alpha = 0.004 \text{ } ^\circ\text{C}^{-1} \pm 1\%$$

$$T = 20 + (6 + 1) \times 5^\circ\text{C} \pm 0.5^\circ\text{C}$$

$$= 55^\circ\text{C} \pm 0.5^\circ\text{C}$$

$$R(T) = R_0 [1 + \alpha(T - 20^\circ\text{C})]$$

$$= 126 [1 + 0.004(55 - 20)]$$

$$= 143.64 \Omega$$

$$\frac{\partial R}{\partial R_0} = 1 + \alpha(T - 20^\circ\text{C})$$

$$= 1 + 0.004(55 - 20)$$

$$= 1.14$$

$$\frac{\partial R}{\partial \alpha} = R_0(T - 20^\circ\text{C})$$

$$= 126(55 - 20)$$

$$= 4410$$

$$\frac{\partial R}{\partial T} = R_0 \alpha$$

$$= 126(0.004)$$

$$= 0.504$$

$$\omega_{R_0} = 126 \times 0.01 = 1.26 \Omega$$

$$\omega_\alpha = 0.004 \times 0.01 = 0.00004 \text{ } ^\circ\text{C}^{-1}$$

$$\omega_T = 55 \times 0.005 = 0.275^\circ\text{C}$$

$$\omega_R = \sqrt{\left(\frac{\partial R}{\partial R_0} \omega_{R_0}\right)^2 + \left(\frac{\partial R}{\partial \alpha} \omega_\alpha\right)^2 + \left(\frac{\partial R}{\partial T} \omega_T\right)^2}$$

$$\omega_R = \sqrt{[1.14(1.26)]^2 + [4410(0.00004)]^2 + [0.504(0.275)]^2}$$

$$= 1.4538 \Omega$$

$$\therefore R_T = 143.64 \pm 1.45 \Omega$$

Assignment 6

$$R_g = 100 \Omega$$

$$I_{fs} = 1 \text{ mA}$$

$$V_{\max} = (26 + 20) \text{ V}$$

$$= 46 \text{ V}$$

$$R_{\text{series}} = \frac{V_{\max}}{I_{fs}} - R_g \rightarrow = 45900 \Omega$$

$$= \frac{46}{1 \times 10^{-3}} - 100$$

Assignment 7

$$N = 8 \text{ bits}$$

$$V = (26/10 + 10) V$$
$$= 12.6 V$$

$$\text{Resolution} = \frac{V_{out}}{2^N}$$
$$= \frac{12.6}{2^8}$$
$$= 0.049 V \text{ per step}$$

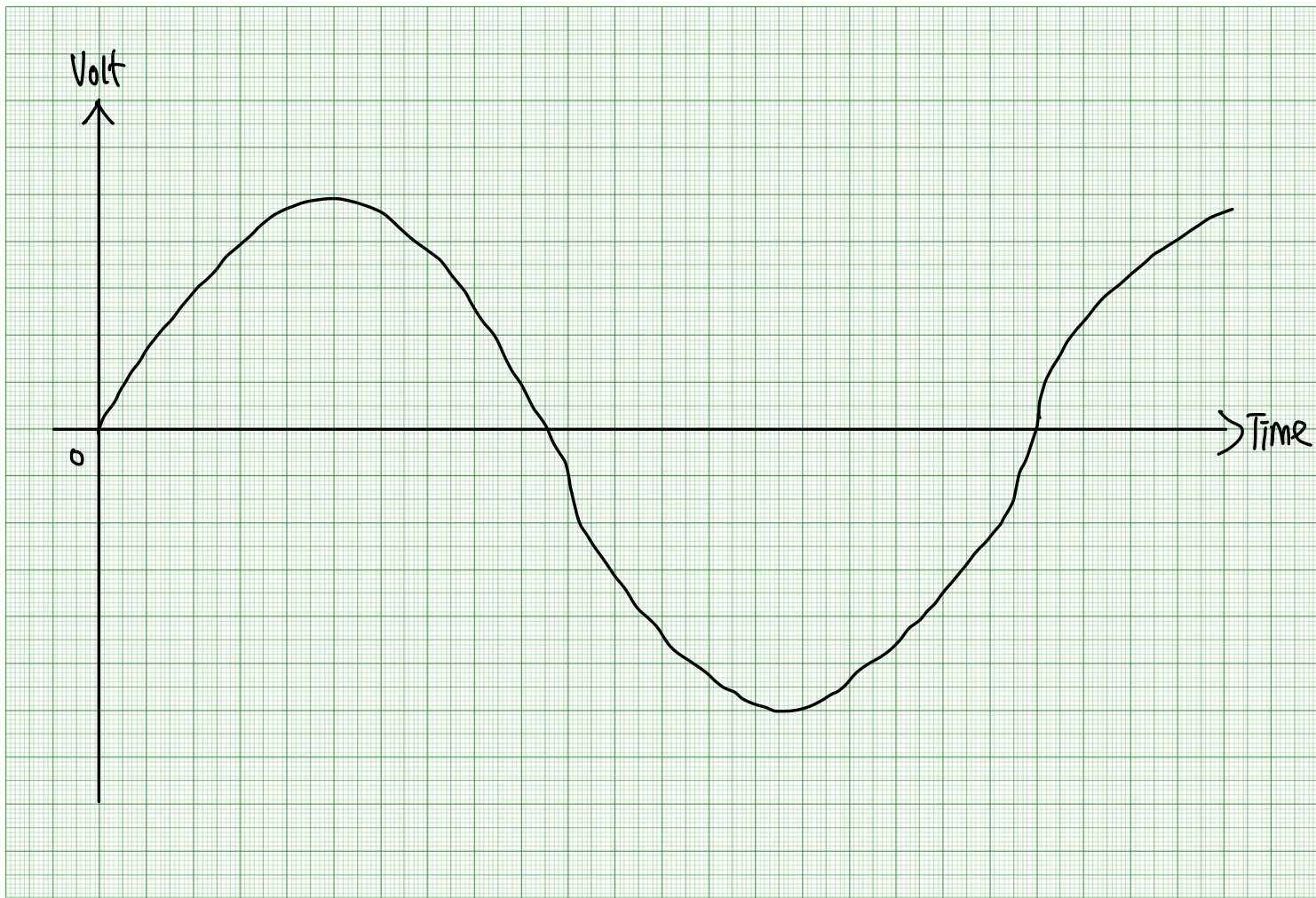
Assignment 8

$$\text{Time base} : (2+6) \times 2 + 1 \text{ ms/div}$$
$$= 17 \text{ ms/div}$$

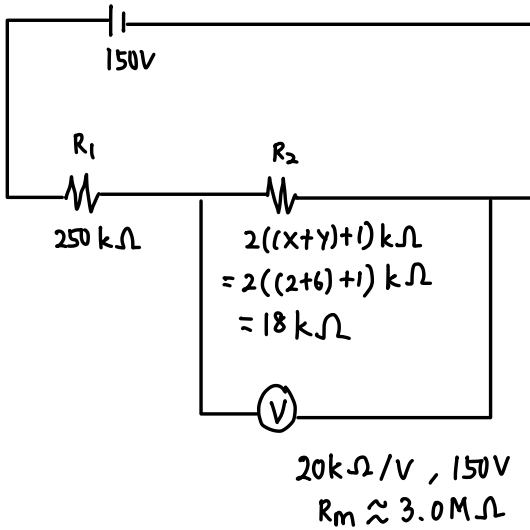
$$\text{full cycle} = (2+6) + 2 \text{ horizontal divisions}$$
$$= 10 \text{ horizontal divisions}$$

$$\text{period} = \text{time base} \times \text{divisions}$$
$$= 17 \times 10$$
$$= 170 \text{ ms/cycle}$$

$$f = \frac{1}{\text{period}}$$
$$= \frac{1}{170 \times 10^{-3}}$$
$$= 5.882 \text{ Hz}$$



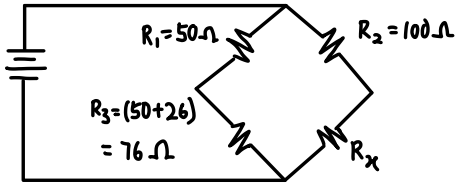
Assignment 9



$$\begin{aligned}
 R_{eq} &= \frac{R_2 R_m}{R_2 + R_m} \\
 &= \frac{18 \times 10^3 (3.0 \times 10^6)}{18 \times 10^3 + 3.0 \times 10^6} \\
 &= 17892.64\ \Omega
 \end{aligned}$$

$$\begin{aligned}
 V &= V_s \left(\frac{R_{eq}}{R_1 + R_{eq}} \right) \\
 &= 150 \left(\frac{17892.64}{250 \times 10^3 + 17892.64} \right) \\
 &= 10.02\text{ V}
 \end{aligned}$$

Assignment 10



$$\begin{aligned}\frac{R_1}{R_2} &= \frac{R_3}{R_x} \\ R_x &= \frac{R_3 \times R_2}{R_1} \\ &= \frac{76 \times 100}{50} \\ &= 152\ \Omega\end{aligned}$$

Assignment 11

$$V_{500} = 18.68\ \text{mV}$$

Cold junction : 0°C

$$\begin{aligned}V_{\text{meas}} &= \frac{26 + 1}{10}\ \text{mV} \\ &= 2.7\ \text{mV}\end{aligned}$$

$$\begin{aligned}\text{Sensitivity} &= \frac{18.68}{500} \\ &= 0.03736\ \text{mV}/^\circ\text{C}\end{aligned}$$

$$\begin{aligned}T &= \frac{V_{\text{meas}}}{V_{500^\circ}} \times 500 \\ &= \frac{2.7}{18.68} \times 500 \\ &= 72.27\ ^\circ\text{C}\end{aligned}$$

Assignment 12

$$A = (26 + 10) \times 10^{-4} \text{ m}^2 = 36 \times 10^{-4} \text{ m}^2$$

$$d = 1.0 \times 10^{-3} \text{ m}$$

$$\epsilon_r = 1.0$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$= \frac{(8.854 \times 10^{-12})(1.0)(36 \times 10^{-4})}{1.0 \times 10^{-3}}$$

$$= 3.187 \times 10^{-11} \text{ F}$$

$$= 31.87 \times 10^{-12} \text{ F}$$

$$= 31.87 \text{ pF}$$

Assignment 13

$$\text{Sensitivity} = 2.00 \text{ V/mm}$$

$$\begin{aligned} \text{Displacement} &= (2 + 6) \times 0.5 \text{ mm} \\ &= 4.5 \text{ mm} \end{aligned}$$

$$\begin{aligned} V_{\text{null}} &= 4.5 \text{ mm} \times 2.00 \text{ V/mm} \\ &= 9.0 \text{ V} \end{aligned}$$

Since the displacement is measured from the null (center) position and linearity is assumed, this value represents the peak output voltage for this displacement.